# PRIVATE CODE ASSISTANT

## Microsoft Global Partner Hackathon - 2026

Team Blue Agents

Capgemini

# PRIVATE CODE ASSISTANT

PROBLEM STATEMENT : Agentic IDEs accelerate code but bypass governance; enterprises need a zero-trust, policy-driven mechanism inside the IDE to enforce standards, security, and ticket adherence at scale.

## OVERVIEW

- ***Rule Drift & Version Fragmentation:*** Governance rules (lint configs, doc templates, commit conventions, allowed tools, prompt policies) often live as copies in many repos or IDE profiles. Teams update them unevenly, so enforcement varies by developer or project. A single, centralized rule pack that the extension reads at runtime eliminates this drift, one update, everyone is current, without asking developers to update the plugin or chase dotfiles.

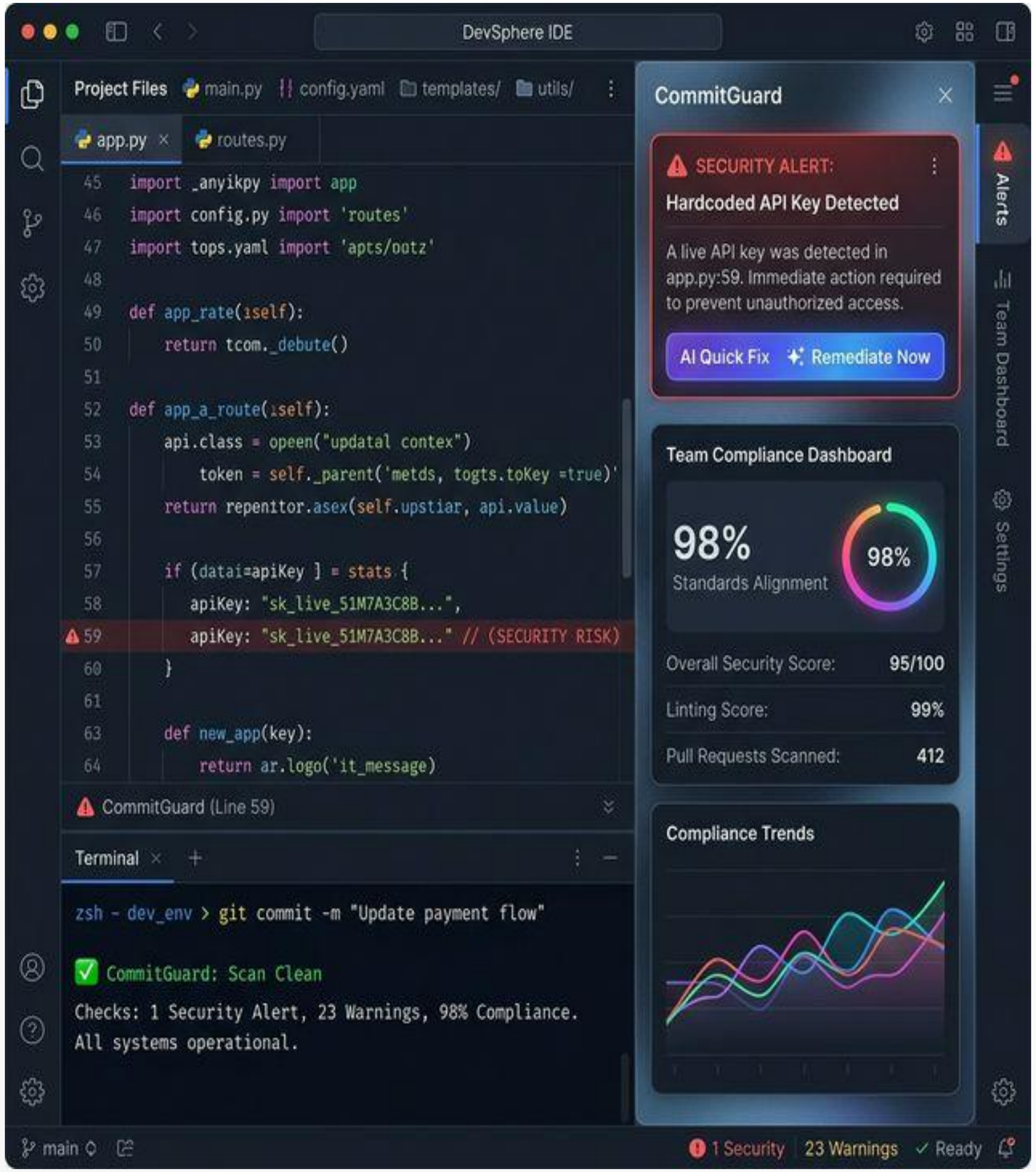
- ***Friction-free Governance :*** Even when rules exist, they're applied only if a developer remembers the correct CLI sequence or installs the right local hooks. Governance that requires manual setup gets bypassed. The extension must auto-apply the latest rules in-editor with minimal clicks, so compliance happens by default, not by discipline.

- ***Scope Creep in Agentic IDE Workflows:*** Agent chats and AI edits inside the IDE can easily diverge from the active ticket/acceptance criteria, creating rework. Developers need ticket-grounded guidance currently of prompting and editing so that the AI and human stay aligned to "the actual ask."

- ***Secret Leaks Before Code Review :*** API keys, cloud creds, and tokens still slip into commits through copy-paste or generated samples. Reviews catch them late, or not at all. Teams need automatic, always-on pre-commit blocking in the editor for secrets and keys, no special setup, so risky content never reaches the repo.

- ***Commit Hygiene & Traceability Gaps :*** Unstructured messages, missing work-item links, and inconsistent branch names break audit trails and change-log automation. The extension should shape commit messages and branches at author time, guided by the active work item, so traceability is guaranteed without extra developer effort.

- ***Silent Policy Drift in Config Files :*** Local changes to policy-sensitive files (e.g., linter rules, tool versions, allowlists) can weaken enforcement without anyone noticing. Developers need immediate, in-editor alerts and guardrails when governance files deviate from the centrally published rule pack.




HACKATHON PROJECT

# Key Features & Functionalities

AI-powered engineering assistant driving faster builds, smarter diagnosis, and automated processes.

## Key Features

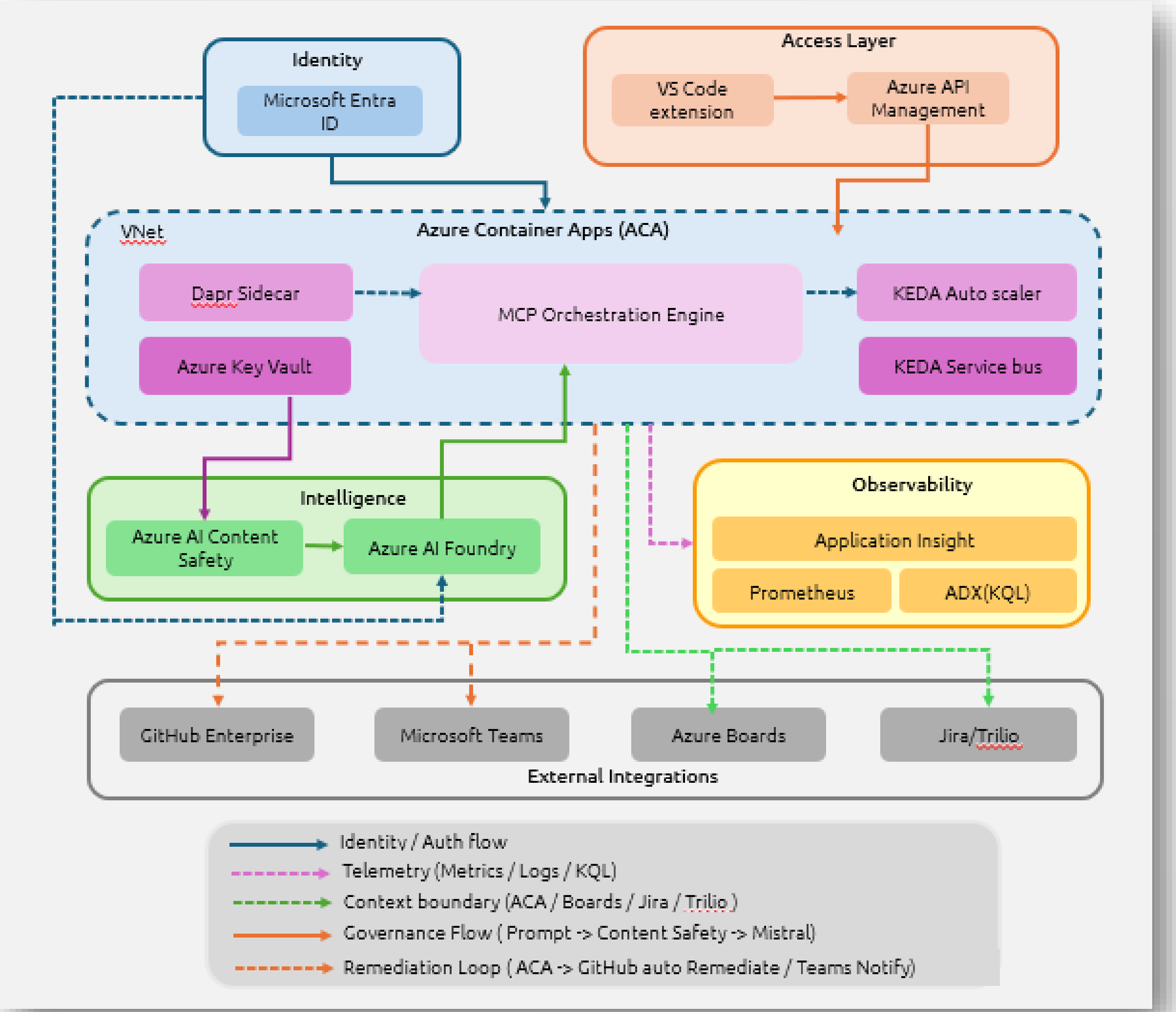
- Zero-Config Bootstrap: One-click environment setup that works instantly across VS Code, Antigravity, and Cursor.
- Enterprise "Golden Rules": Centralized policy management that pushes standards to all developers in real-time.
- AI-Powered Remediation: Uses LLMs to suggest and apply instant fixes for security and naming violations.
- IDE-Agnostic Enforcement: Hooks directly into Git to ensure consistent governance regardless of the editor used.
- Remote Context Sync: Shared task "memory" to ensure seamless developer handovers and model-switching continuity.
- Mistral LLM : train the mistral LLM for Client’s industry-specific terminology, internal code base, arch docs etc.

## Problems Solved

- Secret Leaks: Stops API keys and credentials from ever being committed to the codebase.
- Standards Drift: Eliminates inconsistent naming and architectural patterns across large teams.
- Onboarding Friction: Removes hours of manual tool installation for new developers and AI agents.
- AI Governance: Forces AI coding assistants to adhere to project-specific naming conventions.
- Context Fragmentation: Prevents work from stalling when developers rotate or switch AI models.
- This will improve the code repo maintenance with the unified terminologies and

# Technical Architecture →

This architecture showcases a modular, cloud-native AI orchestration platform built on Azure Container Apps (ACA), integrating identity, intelligence, observability, and external systems into a unified control plane. It provides secure API access, automated remediation, and scalable AI workflows—leveraging Entra ID, Azure AI services, Dapr, KEDA, and robust monitoring to deliver an end-to-end intelligent automation ecosystem.




HACKATHON PROJECT

# Value Adds

AI-powered engineering assistant driving faster builds, smarter diagnosis, and automated processes.

- Fewer CI Failures: Catches errors locally before they hit the build pipeline, saving time and compute costs.
- Instant Developer Readiness: New team members (and AI agents) are compliant and productive on Day 1.
- Seamless Knowledge Handoff: Remote context storage ensures zero downtime when transferring tasks between developers.
- Improved productivity by automation: creation of unit tests, modifying documentation, expediting the manual code reviews, higher quality pull requests
- Automated "Nitpicking": Replaces manual PR comments for style and naming with automated, instant fixes.

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